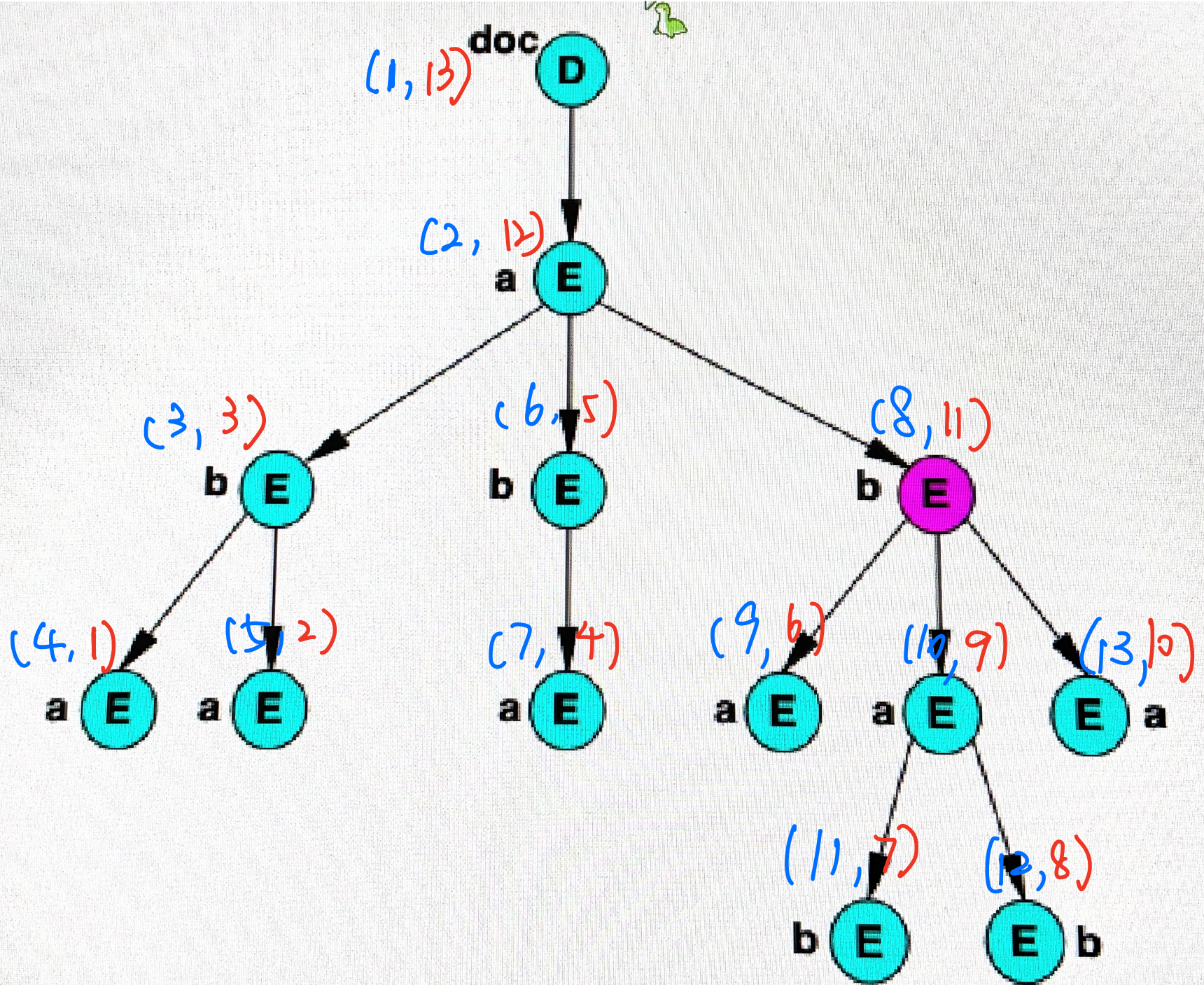




<?xml version="1.0"?>

<a>

<a/>

<a/>

<a/>

<a/>

<a>

<a/>

<a/>

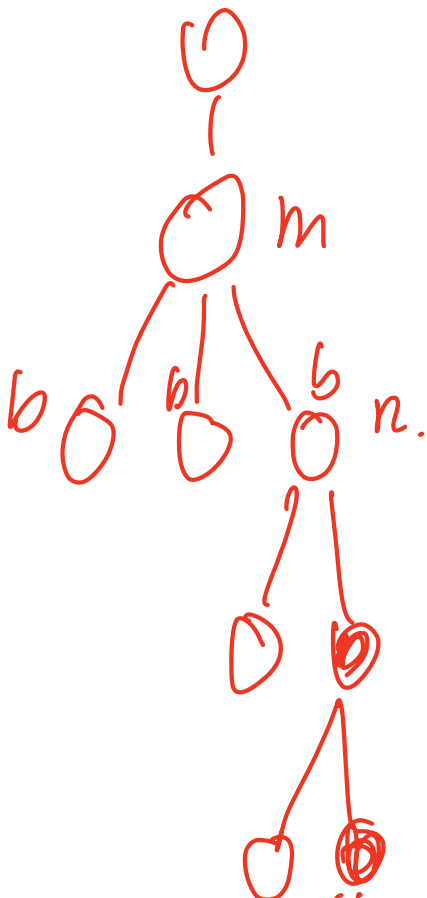
(c) i: the following node a of the
 context node. the node after the end of
 the context node

nothing return.

i: 1, 2, 8.

i: 3, 6.

(d)



// b[3] // * [2].

(10, 2)



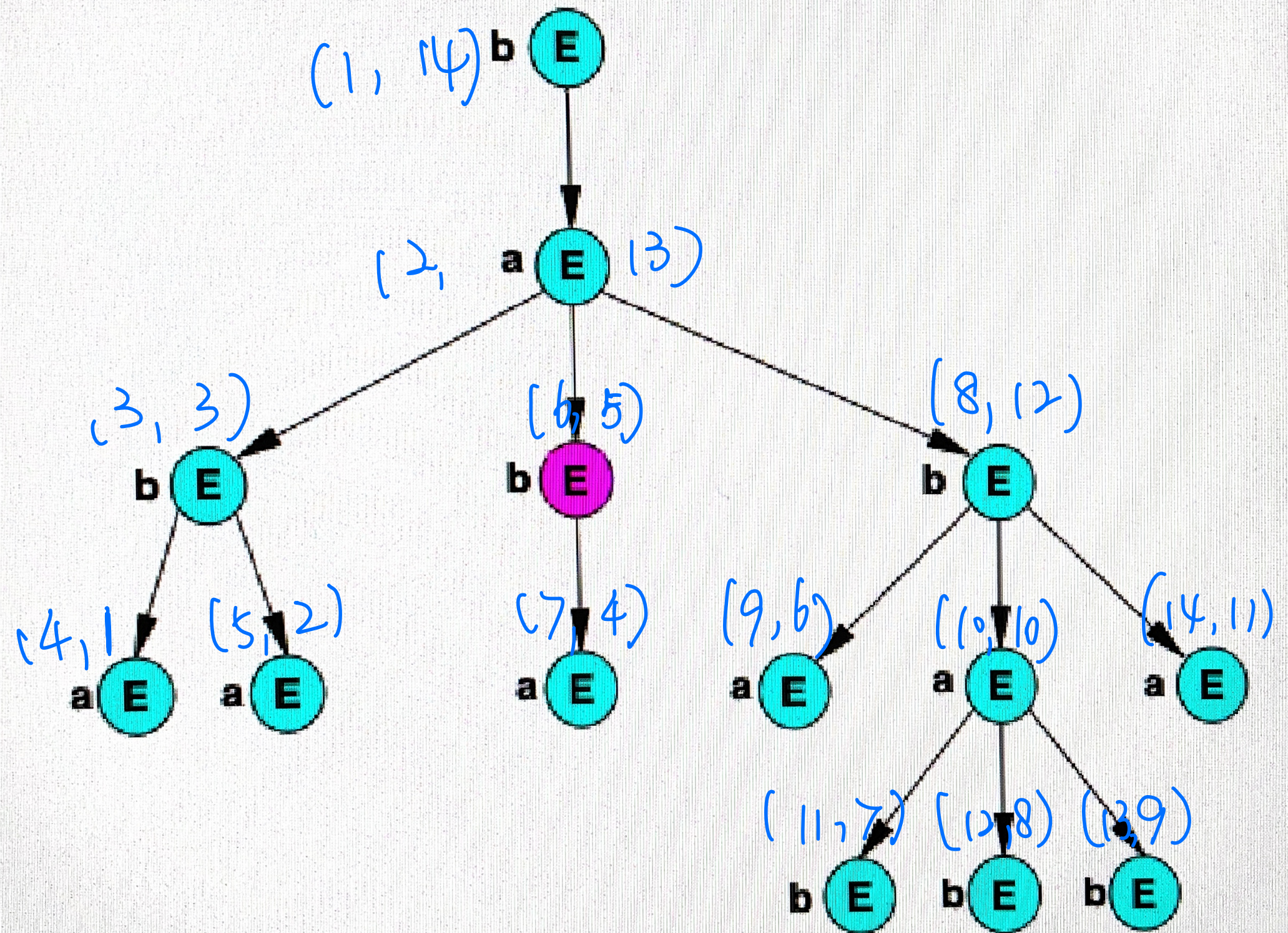
ii:

sibling

// * [counter (pre ceding ^{sibling} :: node) =
counter (following _{sibling} :: node)]

1, 2, 6, 7, 10

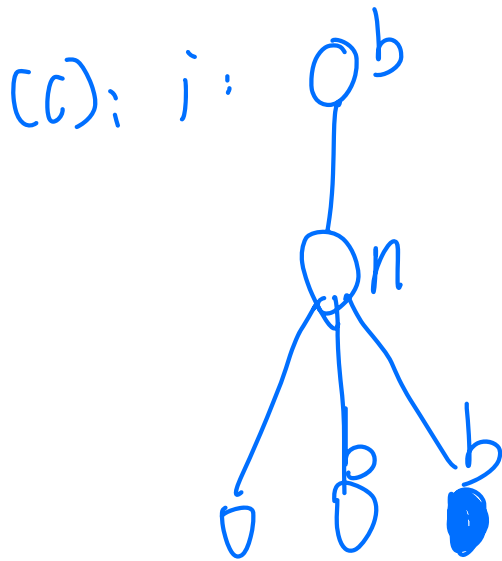
// * [not [*]]



(b): $i: 1, 2,$

$ii: 8, 9, 10, 11, 12, 13, 14$

$iii: 3.$



$//b / * / b [2]$

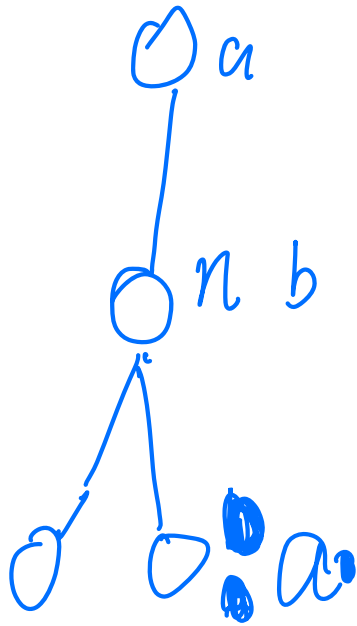
$6, 12,$

$ii:$

$// * [\text{counter} (\text{following} :: *)] =$

$\text{counter} (\text{preceding} :: *)]$

iii:



~~// a / b / a [z]~~

// a / b / * [z] [self::a]

5, 10.

xquery.

for \$m in //trk[difficulty="moderate"]

\$h in //trk[difficulty="hard"]

let \$avg m := avg(\$m/length)

\$avg h := avg(\$h/length)

<return> <difficulty>

<value> { \$m / difficulty } </value>

<tree> { \$m / data (@name) } </tree>

<average length> { \$avg m.

<difficulty>

</return>

Graph database.

$$V: \{1, 2, 3\}$$

$$E: \{10, 11, 12\}$$

$$g: 10 \rightarrow (1, 2)$$

$$11 \rightarrow (2, 3)$$

$$12 \rightarrow (3, 1)$$

$$\lambda: 1 \rightarrow (\text{trek: hut})$$

$$2 \rightarrow (\text{hut})$$

$$3 \rightarrow (\text{hut})$$

$$10 \rightarrow (\text{section})$$

$$11 \rightarrow (\text{section})$$

$$12 \rightarrow (\text{section})$$

$$V : (l, \text{name}) \rightarrow \text{"kepler"}$$

(b)

$$\text{Result}(x) \Leftarrow : \text{trek}(x), : \text{section}(x, y), :$$

$$\text{Section}(y, z), : \text{hnt}(y), : \text{hnt}(z)$$

$$\text{Result}(x) \Leftarrow : \text{trek}(x), : \text{section}(x, y)$$

$$: \text{Section}(y, z), : \text{Section}(z, x),$$

$$: \text{hnt}(y), : \text{hnt}(z), : \text{hnt}(x).$$

(c)

$$\text{result}(x, y) \Leftarrow : : \text{hnt}(x), : \text{Section}(x, y)$$

!hut(y).

x. shower = 'yes'

y. shower = 'yes'

x. #beds ~~15~~ ^{≥ 15}

y. #beds ≥ 15

(D).

result(x) $\leftarrow$:trek(x), :section⁺(x, y)
as e

!hut(y). y. shower = 'yes'
e. distance $\gg 5$
km

(b) Result(x) $\leftarrow$:trek(x), :section(x, y), :
:section(x, y), :hut(y), :hut(x)

Section(y, z), : hnt(y), : hnt(z)

Result(x) $\in$: trek(x), : Section(x, y)

: Section(y, z), : Section(z, x),

: hnt(y), : hnt(z), : hnt(x).

$$\begin{array}{c} \emptyset \\ \searrow \\ \text{src}_1 \end{array} \left(\begin{array}{c} \Phi \\ \searrow \\ \text{src}_1, \text{trg}_2 \end{array} \left(: \text{Section } a, : \text{Section} \right) \cup \begin{array}{c} \psi \\ \searrow \\ \text{src}_1 \end{array} \left(: \right. \right.$$

Section; Section; section)

$$\begin{array}{l} \Phi : \text{trg}_1 = \text{src}_2 \\ \lambda(\text{src}_1) = \text{trek} \\ \lambda(\text{trg}_2) = \text{hnt} \\ \lambda(\text{trg}_1) = \text{hnt}. \end{array}$$

$$\psi : \text{trg}_1 = \text{src}_2$$

$$trg_2 = src_3$$

$$\lambda(src_1) = 'trek'$$

$$\lambda(trg_1) = 'hnt'$$

$$\lambda(src_1) = 'hnt'$$

$$\lambda(src_3) = 'hnt'$$

edge, distance $\in \mathbb{R}$

Map reduce. pagerank,

$$\frac{1}{15} + \frac{1}{5} = \frac{x}{15}$$

$$\frac{1}{10} + \frac{1}{15} = \frac{4}{30} = \frac{2}{15}$$

$$M = \begin{matrix} & \begin{matrix} A & B & C & D & E \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & \frac{1}{3} & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{3} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & 0 \end{bmatrix} \end{matrix} \begin{bmatrix} \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \frac{4}{15} \\ \frac{2}{5} \\ \frac{1}{6} \\ \frac{1}{10} \\ \frac{1}{15} \end{bmatrix}$$

$$\frac{8 + 12 + 5 + 3 + 2}{30} \checkmark$$

map input

Node A: key = $[A, \frac{1}{5}]$, value = $[B]$

B: key = $[B, \frac{1}{5}]$ value = $[C, D]$

C: key = $[C, \frac{1}{5}]$ value = $[B]$

$$D = [D, \frac{1}{5}] \rightsquigarrow [A, C, E]$$

$$E = [E, \frac{1}{5}] \rightsquigarrow [A]$$

map out put.

$$A: [B, \frac{1}{5}], [A, [B]]$$

$$B: [C, \frac{1}{10}], [D, \frac{1}{10}], [B, [C, D]]$$

$$C: [B, \frac{1}{5}], [C, [B]]$$

$$D: [A, \frac{1}{15}], [C, \frac{1}{15}], [E, \frac{1}{15}], [D, [A, C, E]]$$

$$E: [A, \frac{1}{5}], [E, [A]]$$

reduce input

$$A. (A, \frac{1}{15}) (A, \frac{1}{5}) (A, \bar{B})$$

$$B. (B, \frac{1}{5}) (B, \frac{1}{5}) (B, \bar{C}, D)$$

$$C. (C, \frac{1}{10}) (C, \frac{1}{15}) (C, \bar{B})$$

$$D. (D, \frac{1}{10}), (D, \bar{A}, C, E)$$

$$E. (E, \frac{1}{15}), E(\bar{A})$$

Reduce output.

$$A. (A, \frac{4}{15}), (A, \bar{B})$$

$$B. (B, \frac{2}{5}) (B, \bar{C}, D)$$

$$C. (C, \frac{1}{6}) (C, \bar{B})$$

$$D. (D, \frac{1}{10}) (D, \bar{A}, C, E)$$

$$E. (E, \frac{1}{5}) \quad E([A])$$

S has m element
 n bits memory.

k sub array.

$$\binom{k}{11111}$$

$\frac{n}{k}$ burget.

~~$\frac{k}{n}$~~ possible to be
 select

$(1 - \frac{k}{n})$ possible not be
 select

$(1 - \frac{k}{n})^m$ m times not be

select

$$1 - \left(1 - \frac{k}{n}\right)^m$$

select at
least once.

k times

$$\left(1 - \left(1 - \frac{k}{n}\right)^m\right)^k$$

$$\left(1 - \left(1 - \frac{1}{n}\right)^{km}\right)^k$$